

- 1 (a) Two of the SI base quantities and their units are mass (kg) and length (m). Name three other SI base quantities and their units.

1. quantity ..... unit .....

2. quantity ..... unit .....

3. quantity ..... unit .....

[3]

- (b) The pressure  $p$  due to a liquid of density  $\rho$  is related to the depth  $h$  by the expression

$$p = \rho gh,$$

where  $g$  is the acceleration of free fall.

Use this expression to determine the derived units of pressure. Explain your working.

[5]

- 2 A unit is often expressed with a prefix. For example, the gram may be written with the prefix 'kilo' as the kilogram. The prefix represents a power-of-ten. In this case, the power-of-ten is  $10^3$ .

Complete Fig. 1.1 to show each prefix with its symbol and power-of-ten.

| prefix | symbol | power-of-ten |
|--------|--------|--------------|
| kilo   | k      | $10^3$       |
| nano   | n      | .....        |
| centi  | .....  | $10^{-2}$    |
| .....  | M      | $10^6$       |
| .....  | T      | $10^{12}$    |

Fig. 1.1

[4]

- 3 (a) Two of the SI base quantities are mass and time. State three other SI base quantities.

1. ....

2. ....

3. ....

[3]

- (b) A sphere of radius  $r$  is moving at speed  $v$  through air of density  $\rho$ . The resistive force  $F$  acting on the sphere is given by the expression

$$F = Br^2\rho v^k$$

where  $B$  and  $k$  are constants without units.

- (i) State the SI base units of  $F$ ,  $\rho$  and  $v$ .

$F$  .....

$\rho$  .....

$v$  ..... [3]

- (ii) Use base units to determine the value of  $k$ .

$k =$  ..... [2]

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- 4 (a) (i) Distinguish between vector quantities and scalar quantities.

.....  
 .....  
 ..... [2]

- (ii) State whether each of the following is a vector quantity or a scalar quantity.

1. temperature

..... [1]

2. acceleration of free fall

..... [1]

3. electrical resistance

..... [1]

- (b) A block of wood of weight 25 N is held stationary on a slope by means of a string, as shown in Fig. 1.1.

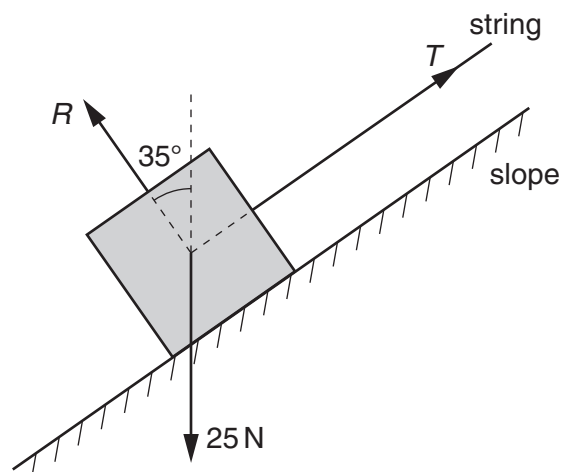


Fig. 1.1

The tension in the string is  $T$  and the slope pushes on the block with a force  $R$  that is normal to the slope.

Either by scale drawing on Fig. 1.1 or by calculation, determine the tension  $T$  in the string.

$T = \dots\dots\dots$  N [3]

- 5 (a) Distinguish between *scalar* quantities and *vector* quantities.

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.....  
 .....  
 ..... [2]

- (b) In the following list, underline **all** the scalar quantities.

acceleration      force      kinetic energy      mass      power      weight      [1]

- 6 (a) (i) State the SI base units of volume.

base units of volume ..... [1]

- (ii) Show that the SI base units of pressure are  $\text{kg m}^{-1} \text{s}^{-2}$ .

[1]

- (b) The volume  $V$  of liquid that flows through a pipe in time  $t$  is given by the equation

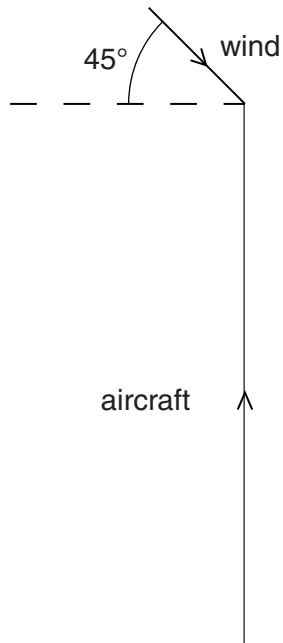
$$\frac{V}{t} = \frac{\pi P r^4}{8 C l}$$

where  $P$  is the pressure difference between the ends of the pipe of radius  $r$  and length  $l$   
 The constant  $C$  depends on the frictional effects of the liquid.  
 Determine the base units of  $C$ .

base units of  $C$  ..... [3]

---

- 7 (e) The velocity vector diagram for an aircraft heading due north is shown to scale in Fig. 1.1. There is a wind blowing from the north-west.



**Fig. 1.1**

The speed of the wind is  $36 \text{ m s}^{-1}$  and the speed of the aircraft is  $250 \text{ m s}^{-1}$ .

- (i) Draw an arrow on Fig. 1.1 to show the direction of the resultant velocity of the aircraft. [1]
- (ii) Determine the magnitude of the resultant velocity of the aircraft.

resultant velocity = .....  $\text{m s}^{-1}$  [2]

---

- 8 (a) State the SI base units of force.

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.....[1]

- (b) Two wires each of length  $l$  are placed parallel to each other a distance  $x$  apart, as shown in Fig. 1.1.

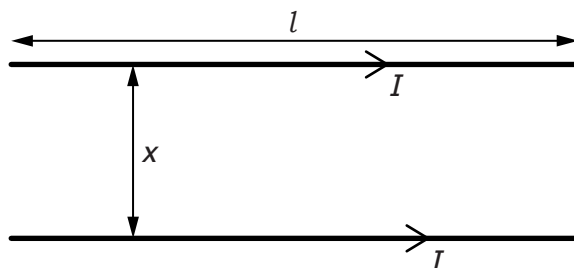


Fig. 1.1

Each wire carries a current  $I$ . The currents give rise to a force  $F$  on each wire given by

$$F = \frac{KI^2l}{x}$$

where  $K$  is a constant.

- (i) Determine the SI base units of  $K$ .

units of  $K$ ..... [2]

- (ii) On Fig. 1.2, sketch the variation with  $x$  of  $F$ . The quantities  $I$  and  $l$  remain constant.



Fig. 1.2

[2]

- (iii) The current  $I$  in both of the wires is varied.

On Fig. 1.3, sketch the variation with  $I$  of  $F$ . The quantities  $x$  and  $l$  remain constant.



Fig. 1.3

[1]

- 9 (a) State two SI base units other than the kilogram, metre and second. 9702/22/O/N/13/Q1

1. ....

2. .... [2]

- (b) A metal wire has original length  $l_0$ . It is then suspended and hangs vertically as shown in Fig. 1.1.

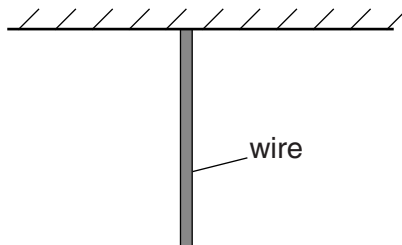


Fig. 1.1

The weight of the wire causes it to stretch. The elastic potential energy stored in the wire is  $E$ .

- (i) Show that the SI base units of  $E$  are  $\text{kg m}^2 \text{s}^{-2}$ .

[2]

- (ii) The elastic potential energy  $E$  is given by

$$E = C\rho^2 g^2 A l_0^3$$

where  $\rho$  is the density of the metal,  
 $g$  is the acceleration of free fall,  
 $A$  is the cross-sectional area of the wire  
and  $C$  is a constant.

Determine the SI base units of  $C$ .

SI base units of  $C$  ..... [3]

**10 (a)** Show that the SI base units of power are  $\text{kg m}^2 \text{s}^{-3}$ .

**9702/22/M/J/14/Q1**

[3]

**(b)** The rate of flow of thermal energy  $\frac{Q}{t}$  in a material is given by

$$\frac{Q}{t} = \frac{CAT}{x}$$

where  $A$  is the cross-sectional area of the material,  
 $T$  is the temperature difference across the thickness of the material,  
 $x$  is the thickness of the material,  
 $C$  is a constant.

Determine the SI base units of  $C$ .

base units ..... [4]

---

**11 (a)** Underline **all** the base quantities in the following list.

**9702/23/M/J/14/Q1**

ampere    charge    current    mass    second    temperature    weight    [2]

**(b)** The potential energy  $E_p$  stored in a stretched wire is given by

$$E_p = \frac{1}{2} C \sigma^2 V$$

where  $C$  is a constant,  
 $\sigma$  is the strain,  
 $V$  is the volume of the wire.

Determine the SI base units of  $C$ .

base units ..... [3]

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- 12 (a) Mass, length and time are SI base quantities.  
State two other base quantities.

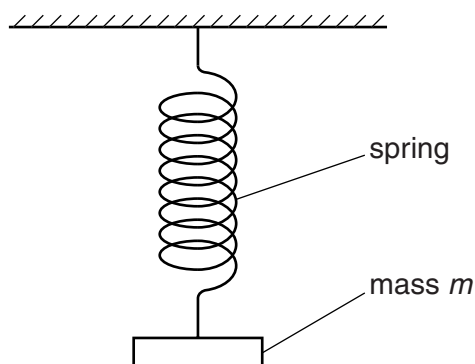
9702/21/O/N/14/Q1

1. ....

2. ....

[2]

- (b) A mass  $m$  is placed on the end of a spring that is hanging vertically, as shown in Fig. 1.1.



**Fig. 1.1**

The mass is made to oscillate vertically. The time period of the oscillations of the mass is  $T$ .  
The period  $T$  is given by

$$T = C \sqrt{\frac{m}{k}}$$

where  $C$  is a constant and  $k$  is the spring constant.  
Show that  $C$  has no units.

[3]

- 13 (c) An object B is on a horizontal surface. Two forces act on B in this horizontal plane. A vector diagram for these forces is shown to scale in Fig. 1.1.

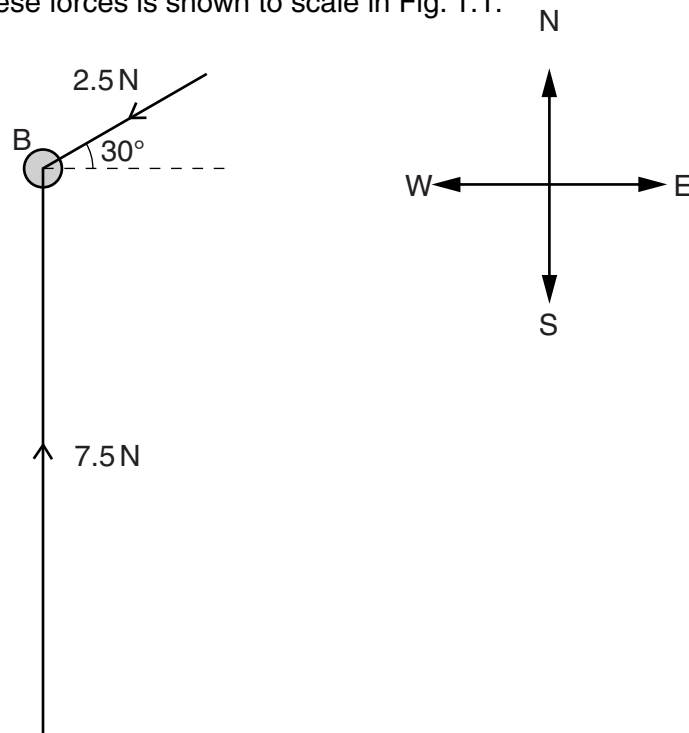


Fig. 1.1

A force of 7.5 N towards north and a force of 2.5 N from 30° north of east act on B. The mass of B is 750 g.

- (i) On Fig. 1.1, draw an arrow to show the approximate direction of the resultant of these two forces. [1]

- (ii) 1. Show that the magnitude of the resultant force on B is 6.6 N. [1]

2. Calculate the magnitude of the acceleration of B produced by this resultant force.

magnitude = .....  $\text{ms}^{-2}$  [2]

- (iii) Determine the angle between the direction of the acceleration and the direction of the 7.5 N force.

angle = ..... ° [1]

14 (a) Force is a vector quantity. State three other vector quantities.

9702/23/O/N/14/Q3

1. ....
2. ....
3. ....

[2]

(b) Three coplanar forces  $X$ ,  $Y$  and  $Z$  act on an object, as shown in Fig. 3.1.

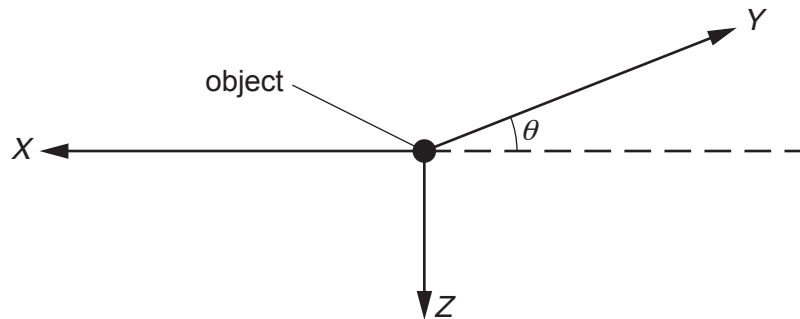


Fig. 3.1

The force  $Z$  is vertical and  $X$  is horizontal. The force  $Y$  is at an angle  $\theta$  to the horizontal. The force  $Z$  is kept constant at 70 N.

In an experiment, the magnitude of force  $X$  is varied. The magnitude and direction of force  $Y$  are adjusted so that the object remains in equilibrium.

Fig. 3.2 shows the variation of the magnitude of force  $Y$  with the magnitude of force  $X$ .

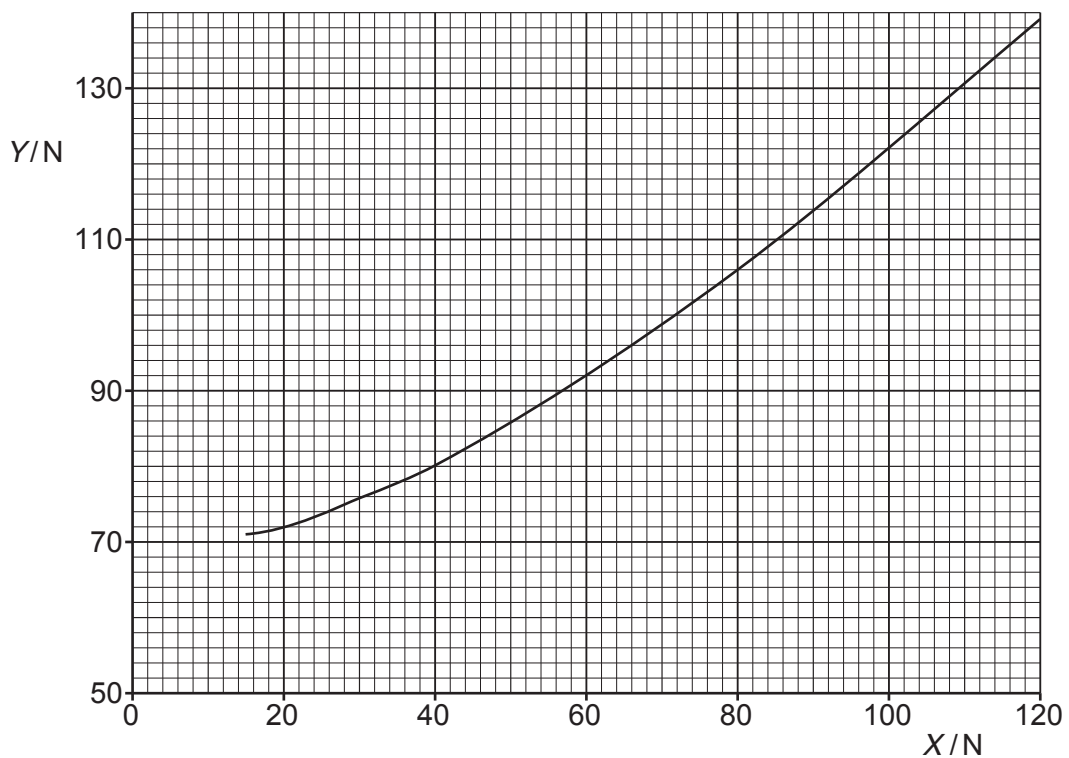


Fig. 3.2

- (i) Use Fig. 3.2 to estimate the magnitude of  $Y$  for  $X = 0$ .

$Y =$  ..... N [1]

- (ii) State and explain the value of  $\theta$  for  $X = 0$ .

.....  
.....  
..... [2]

- (iii) The magnitude of  $X$  is increased to 160 N. Use resolution of forces to calculate the value of

1. angle  $\theta$ ,

$\theta =$  ..... ° [2]

2. the magnitude of force  $Y$ .

$Y =$  ..... N [2]

- (c) The angle  $\theta$  decreases as  $X$  increases. Explain why the object cannot be in equilibrium for  $\theta = 0$ .

.....  
.....  
..... [1]

---

- 15 (a)** The kilogram, metre and second are SI base units.  
State two other base units.

1. ....

2. ....

- (b)** Determine the SI base units of [2]

**(i)** stress,

**(ii)** the Young modulus. SI base units ..... [2]

SI base units ..... [1]

- 16 (a)** Use the definition of power to show that the SI base units of power are  $\text{kg m}^2 \text{s}^{-3}$ .

[2]

- (b)** Use an expression for electrical power to determine the SI base units of potential difference.

units ..... [2]

- 17 (a)** Use the definition of work done to show that the SI base units of energy are  $\text{kg m}^2 \text{s}^{-2}$ .

[2]

- (b)** Define potential difference.

.....

..... [1]

- (c)** Determine the SI base units of resistance. Show your working.

units ..... [3]

- 18 (a) The distance between the Sun and the Earth is  $1.5 \times 10^{11}$  m. State this distance in Gm.

distance = ..... Gm [1]

- (b) The distance from the centre of the Earth to a satellite above the equator is 42.3 Mm. The radius of the Earth is 6380 km.  
A microwave signal is sent from a point on the Earth directly below the satellite.  
Calculate the time taken for the microwave signal to travel to the satellite and back.

time = ..... s [2]

- (c) The speed  $v$  of a sound wave through a gas of density  $\rho$  and pressure  $P$  is given by

$$v = \sqrt{\frac{CP}{\rho}}$$

where  $C$  is a constant.

Show that  $C$  has no unit.

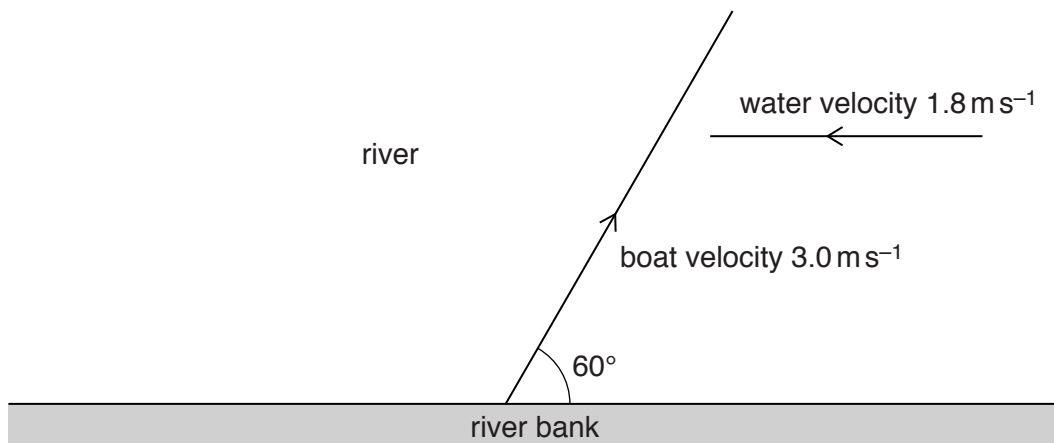
[3]

- (d) Underline all the scalar quantities in the list below.

acceleration      energy      momentum      power      weight

[1]

- (e) A boat travels across a river in which the water is moving at a speed of  $1.8 \text{ m s}^{-1}$ . The velocity vectors for the boat and the river water are shown to scale in Fig. 1.1.



**Fig. 1.1** (shown to scale)

In still water the speed of the boat is  $3.0 \text{ m s}^{-1}$ . The boat is directed at an angle of  $60^\circ$  to the river bank.

- (i) On Fig. 1.1, draw a vector triangle or a scale diagram to show the resultant velocity of the boat.

[2]

- (ii) Determine the magnitude of the resultant velocity of the boat.

resultant velocity = .....  $\text{m s}^{-1}$  [2]

---

**19 (a)** A list of quantities that are either scalars or vectors is shown in Fig. 1.1.

**9702/23/M/J/16/Q1**

| quantity | scalar | vector |
|----------|--------|--------|
| distance | ✓      |        |
| energy   |        |        |
| momentum |        |        |
| power    |        |        |
| time     |        |        |
| weight   |        |        |

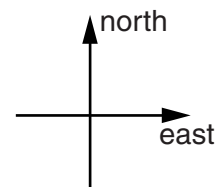
**Fig. 1.1**

Complete Fig. 1.1 to indicate whether each quantity is a scalar or a vector.  
One line has been completed as an example.

[2]

**(b)** A girl runs 120 m due north in 15 s. She then runs 80 m due east in 12 s.

- (i)** Sketch a vector diagram to show the path taken by the girl. Draw and label her resultant displacement R.



[1]

(ii) Calculate, for the girl,

1. the average speed,

average speed = .....  $\text{ms}^{-1}$  [1]

2. the magnitude of the average velocity  $v$  and its angle with respect to the direction of the initial path.

magnitude of  $v$  = .....  $\text{ms}^{-1}$

angle = ..... $^{\circ}$   
[3]

---

20 (a) (i) Define *pressure*.

9702/22/O/N/16/Q1

.....  
.....[1]

(ii) Show that the SI base units of pressure are  $\text{kg m}^{-1} \text{s}^{-2}$ .

[1]

- (b) Gas flows through the narrow end (nozzle) of a pipe. Under certain conditions, the mass  $m$  of gas that flows through the nozzle in a short time  $t$  is given by

$$\frac{m}{t} = kC \sqrt{\rho P}$$

where  $k$  is a constant with no units,  
 $C$  is a quantity that depends on the nozzle size,  
 $\rho$  is the density of the gas arriving at the nozzle,  
 $P$  is the pressure of the gas arriving at the nozzle.  
 Determine the base units of  $C$ .

base units .....[3]

- 
- 21 (a) State two SI base units other than kilogram, metre and second. 9702/22/M/J/17/Q1

1. ....

2. ....

[1]

- (b) Determine the SI base units of resistivity.

base units .....[3]

---

- 22 The drag force  $F_D$  acting on a sphere moving through a fluid is given by the expression

$$F_D = K\rho v^2$$

where  $K$  is a constant,  
 $\rho$  is the density of the fluid  
 and  $v$  is the speed of the sphere.  
 Determine the SI base units of  $K$ .

base units .....[3]

---

- 23 (a) (i) Define *power*.

.....  
 .....[1]

- (ii) Show that the SI base units of power are  $\text{kg m}^2 \text{s}^{-3}$ .

- (b) All bodies radiate energy. The power  $P$  radiated by a body is given by

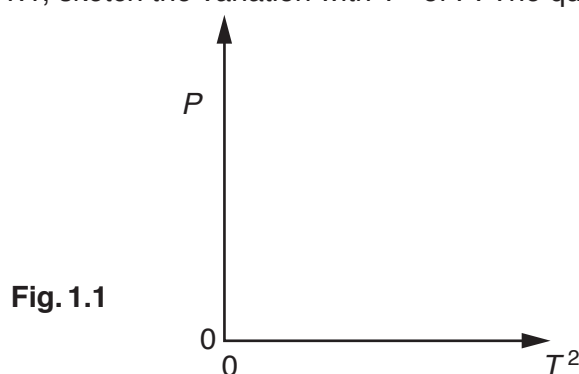
$$P = kAT^4$$

where  $T$  is the thermodynamic temperature of the body,  
 $A$  is the surface area of the body  
and  $k$  is a constant.

- (i) Determine the SI base units of  $k$ .

base units .....[2]

- (ii) On Fig. 1.1, sketch the variation with  $T^2$  of  $P$ . The quantity  $A$  remains constant.



[1]

- 24 (a)** State what is meant by a *scalar* quantity and by a *vector* quantity.

**9702/21/M/J/18/Q1**

scalar:.....

.....

vector:.....

.....

[2]

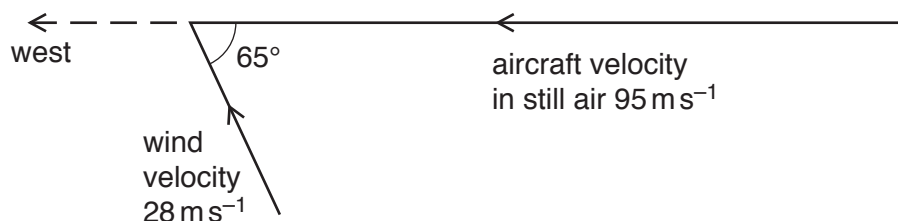
- (b) Complete Fig. 1.1 to indicate whether each of the quantities is a vector or a scalar.

| quantity    | vector or scalar |
|-------------|------------------|
| power       |                  |
| temperature |                  |
| momentum    |                  |

**Fig. 1.1**

[2]

- (c) An aircraft is travelling in wind. Fig. 1.2 shows the velocities for the aircraft in still air and for the wind.



**Fig. 1.2**

The velocity of the aircraft in still air is  $95 \text{ m s}^{-1}$  to the west.  
The velocity of the wind is  $28 \text{ m s}^{-1}$  from  $65^\circ$  south of east.

- (i) On Fig. 1.2, draw an arrow, labelled R, in the direction of the resultant velocity of the aircraft. [1]
- (ii) Determine the magnitude of the resultant velocity of the aircraft.

magnitude of velocity = .....  $\text{m s}^{-1}$  [2]

---

25 (a) Define *force*.

.....[1]

(b) State the SI base units of force.

.....[1]

(c) The force  $F$  between two point charges is given by  $F = \frac{Q_1 Q_2}{4\pi r^2 \epsilon}$

where  $Q_1$  and  $Q_2$  are the charges,

$r$  is the distance between the charges,

$\epsilon$  is a constant that depends on the medium between the charges.

Use the above expression to determine the base units of  $\epsilon$ .

base units .....[2]

---

26 (a) (i) Define *resistance*.

.....  
 .....[1]

(ii) A potential difference of 0.60 V is applied across a resistor of resistance 4.0 GΩ.

Calculate the current, in pA, in the resistor.

current = ..... pA [2]

(b) The energy  $E$  transferred when charge  $Q$  moves through an electrical component is given by the equation

$$E = QV$$

where  $V$  is the potential difference across the component.

Use the equation to determine the SI base units of potential difference.

SI base units ..... [3]

---

- 27 (a) Distinguish between vector and scalar quantities.

.....  
 .....  
 ..... [2]

- (b) The electric field strength  $E$  at a distance  $x$  from an isolated point charge  $Q$  is given by the equation

$$E = \frac{Q}{x^2 b}$$

where  $b$  is a constant.

- (i) Use the definition of electric field strength to show that  $E$  has SI base units of  $\text{kg m A}^{-1} \text{s}^{-3}$ .

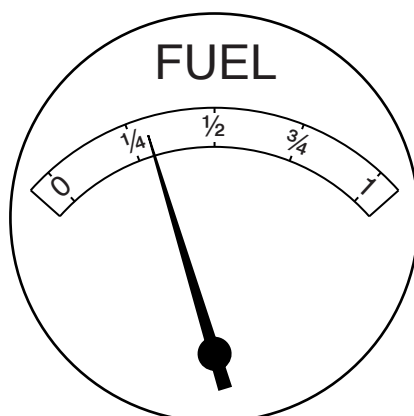
[2]

- (ii) Use the units for  $E$  given in (b)(i) to determine the SI base units of  $b$ .

SI base units of  $b$  ..... [2]

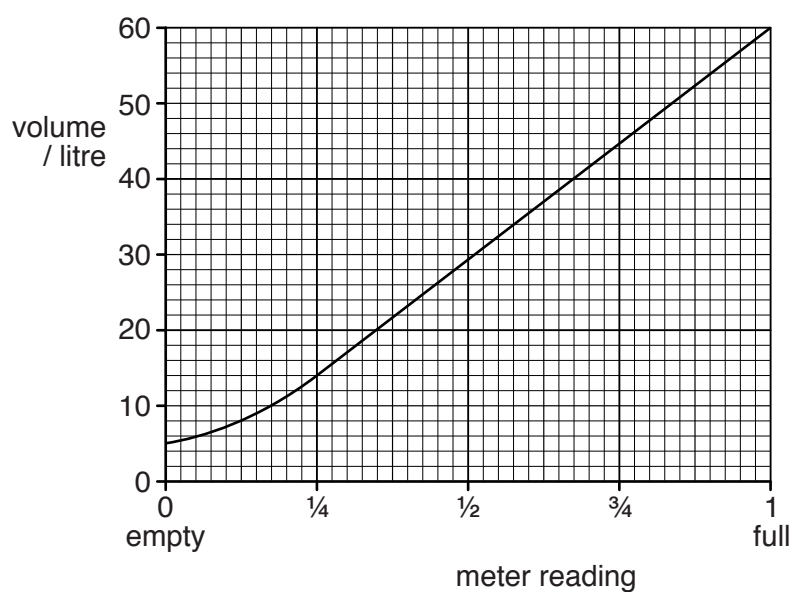
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**Hy** The volume of fuel in the tank of a car is monitored using a meter as illustrated in Fig. 1.1.



**Fig. 1.1**

The meter has an analogue scale. The meter reading for different volumes of fuel in the tank is shown in Fig. 1.2.



**Fig. 1.2**

The meter is calibrated in terms of the fraction of the tank that remains filled with fuel.

**(a)** The car uses 1.0 litre of fuel when travelling 14 km. The car starts a journey with a full tank of fuel.

**(i)** Calculate the volume of fuel remaining in the tank after a journey of 210 km.

volume = ..... litres [2]

**(ii)** Use your answer to **(i)** and Fig. 1.2 to determine the change in the meter reading during the 210 km journey.

from *full* to ..... [1]

**(b)** There is a systematic error in the meter.

**(i)** State the feature of Fig. 1.2 that indicates that there is a systematic error.

.....  
..... [1]

**(ii)** Suggest why, for this meter, it is an advantage to have this systematic error.

.....  
..... [1]

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**Hφ** A simple pendulum may be used to determine a value for the acceleration of free fall  $g$ . Measurements are made of the length  $L$  of the pendulum and the period  $T$  of oscillation. The values obtained, with their uncertainties, are as shown.

$$T = (1.93 \pm 0.03) \text{ s}$$

$$L = (92 \pm 1) \text{ cm}$$

**(a)** Calculate the percentage uncertainty in the measurement of

**(i)** the period  $T$ ,

uncertainty = ..... % [1]

**(ii)** the length  $L$ .

uncertainty = ..... % [1]

**(b)** The relationship between  $T$ ,  $L$  and  $g$  is given by

$$g = \frac{4\pi^2 L}{T^2}.$$

Using your answers in **(a)**, calculate the percentage uncertainty in the value of  $g$ .

uncertainty = ..... % [1]

**(c)** The values of  $L$  and  $T$  are used to calculate a value of  $g$  as  $9.751 \text{ ms}^{-2}$ .

**(i)** By reference to the measurements of  $L$  and  $T$ , suggest why it would not be correct to quote the value of  $g$  as  $9.751 \text{ ms}^{-2}$ .

.....

..... [1]

**(ii)** Use your answer in **(b)** to determine the absolute uncertainty in  $g$ .

Hence state the value of  $g$ , with its uncertainty, to an appropriate number of significant figures.

$g = \dots \pm \dots \text{ ms}^{-2}$  [2]

---

**30** A metal wire has a cross-section of diameter approximately 0.8 mm.

**(a)** State what instrument should be used to measure the diameter of the wire.

.....[1]

**(b)** State how the instrument in **(a)** is

**(i)** checked so as to avoid a systematic error in the measurements,

.....  
 .....[1]

**(ii)** used so as to reduce random errors.

.....  
 .....  
 .....[2]

**31** Measurements made for a sample of metal wire are shown in Fig. 1.1. **9702/21/M/J/11/Q1**

| quantity   | measurement  | uncertainty      |
|------------|--------------|------------------|
| length     | 1750 mm      | $\pm 3$ mm       |
| diameter   | 0.38 mm      | $\pm 0.01$ mm    |
| resistance | $7.5 \Omega$ | $\pm 0.2 \Omega$ |

**Fig. 1.1**

**(a)** State the appropriate instruments used to make each of these measurements.

**(i)** length ..... [1]

**(ii)** diameter ..... [1]

**(iii)** resistance ..... [1]

**(b)** **(i)** Show that the resistivity of the metal is calculated to be  $4.86 \times 10^{-7} \Omega \text{ m}$ .

[2]

(ii) Calculate the uncertainty in the resistivity.

uncertainty =  $\pm$  .....  $\Omega \text{ m}$  [4]

(c) Use the answers in (b) to express the resistivity with its uncertainty to the appropriate number of significant figures.

resistivity = .....  $\pm$  .....  $\Omega \text{ m}$  [1]

9702/22/M/J/12/Q1

32 The volume  $V$  of liquid flowing in time  $t$  through a pipe of radius  $r$  is given by the equation

$$\frac{V}{t} = \frac{\pi P r^4}{8 C l}$$

where  $P$  is the pressure difference between the ends of the pipe of length  $l$ , and  $C$  depends on the frictional effects of the liquid.

An experiment is performed to determine  $C$ . The measurements made are shown in Fig. 1.1.

Fig. 1.1

| $\frac{V}{t} / 10^{-6} \text{ m}^3 \text{ s}^{-1}$ | $P / 10^3 \text{ N m}^{-2}$ | $r / \text{ mm}$ | $l / \text{ m}$   |
|----------------------------------------------------|-----------------------------|------------------|-------------------|
| $1.20 \pm 0.01$                                    | $2.50 \pm 0.05$             | $0.75 \pm 0.01$  | $0.250 \pm 0.001$ |

(a) Calculate the value of  $C$ .

$C =$  .....  $\text{ N s m}^{-2}$  [2]

(b) Calculate the uncertainty in  $C$ .

uncertainty = .....  $\text{Ns m}^{-2}$  [3]

(c) State the value of  $C$  and its uncertainty to the appropriate number of significant figures.

$C = \dots\dots\dots \pm \dots\dots\dots \text{Ns m}^{-2}$  [1]

---

**9702/23/O/N/12/Q1**

**33** (a) The spacing between two atoms in a crystal is  $3.8 \times 10^{-10} \text{ m}$ . State this distance in pm.

spacing = ..... pm [1]

(b) Calculate the time of one day in Ms.

time = ..... Ms [1]

(c) The distance from the Earth to the Sun is 0.15 Tm. Calculate the time in minutes for light to travel from the Sun to the Earth.

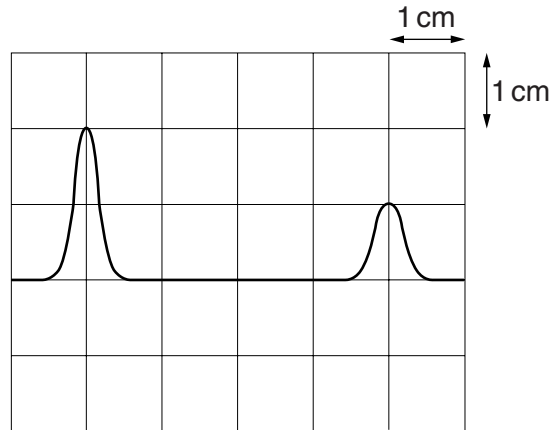
time = ..... min [2]

(d) Underline all the vector quantities in the list below.

|          |        |          |        |      |     |
|----------|--------|----------|--------|------|-----|
| distance | energy | momentum | weight | work | [1] |
|----------|--------|----------|--------|------|-----|

---

- 34 A source of radio waves sends a pulse towards a reflector. The pulse returns from the reflector and is detected at the same point as the source. The emitted and reflected pulses are recorded on a cathode-ray oscilloscope (c.r.o.) as shown in Fig. 2.1.



**Fig. 2.1**

The time-base setting is  $0.20 \mu\text{s cm}^{-1}$ .

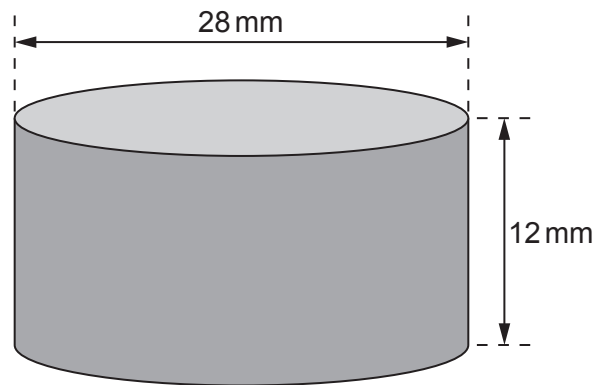
- (a) Using Fig. 2.1, determine the distance between the source and the reflector.

distance = ..... m [4]

- (b) Determine the time-base setting required to produce the same separation of pulses on the c.r.o. when sound waves are used instead of radio waves.  
The speed of sound is  $300 \text{ m s}^{-1}$ .

.....  
 .....  
 ..... [3]

- 35 A cylindrical disc is shown in Fig. 1.1.



**Fig. 1.1**

The disc has diameter 28 mm and thickness 12 mm.  
The material of the disc has density  $6.8 \times 10^3 \text{ kg m}^{-3}$ .

Calculate, to two significant figures, the weight of the disc.

weight = ..... N [4]

---

- 36** The time  $T$  for a satellite to orbit the Earth is given by

$$T = \sqrt{\left(\frac{KR^3}{M}\right)}$$

where  $R$  is the distance of the satellite from the centre of the Earth,  
 $M$  is the mass of the Earth,  
 and  $K$  is a constant.

- (a)** Determine the SI base units of  $K$ .

SI base units of  $K$  ..... [2]

- (b)** Data for a particular satellite are given in Fig. 2.1.

| quantity | measurement                     | uncertainty |
|----------|---------------------------------|-------------|
| $T$      | $8.64 \times 10^4 \text{ s}$    | $\pm 0.5\%$ |
| $R$      | $4.23 \times 10^7 \text{ m}$    | $\pm 1\%$   |
| $M$      | $6.0 \times 10^{24} \text{ kg}$ | $\pm 2\%$   |

**Fig. 2.1**

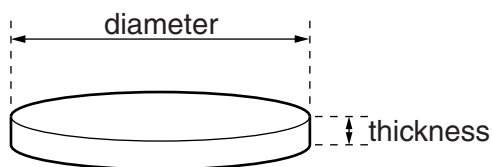
Calculate  $K$  and its actual uncertainty in SI units.

$K = \dots \pm \dots$  SI units [4]

---

37 A coin is made in the shape of a thin cylinder, as shown in Fig. 2.1.

9702/22/M/J/14/Q2



**Fig. 2.1**

Fig. 2.2 shows the measurements made in order to determine the density  $\rho$  of the material used to make the coin.

| quantity  | measurement | uncertainty   |
|-----------|-------------|---------------|
| mass      | 9.6 g       | $\pm 0.5$ g   |
| thickness | 2.00 mm     | $\pm 0.01$ mm |
| diameter  | 22.1 mm     | $\pm 0.1$ mm  |

**Fig. 2.2**

(a) Calculate the density  $\rho$  in  $\text{kg m}^{-3}$ .

$$\rho = \dots\dots\dots \text{kg m}^{-3} \quad [3]$$

(b) (i) Calculate the percentage uncertainty in  $\rho$ .

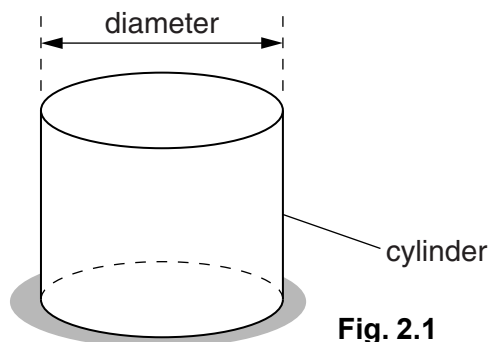
$$\text{percentage uncertainty} = \dots\dots\dots [3]$$

(ii) State the value of  $\rho$  with its actual uncertainty.

$$\rho = \dots\dots\dots \pm \dots\dots\dots \text{kg m}^{-3} \quad [1]$$

..... [1]

- (b) A cylinder is placed on a horizontal surface, as shown in Fig. 2.1.



The following measurements were made on the cylinder:

mass =  $5.09 \pm 0.01$  kg

diameter =  $9.4 \pm 0.1$  cm.

- (i) Calculate the pressure produced by the cylinder on the surface.

pressure = ..... Pa [3]

- (ii) Calculate the actual uncertainty in the pressure.

actual uncertainty = ..... Pa [3]

- (iii) State the pressure, with its actual uncertainty.

pressure = .....  $\pm$  ..... Pa [1]

---

- 39 The resistance  $R$  of a uniform metal wire is measured for different lengths  $l$  of the wire. The variation with  $l$  of  $R$  is shown in Fig. 3.1.

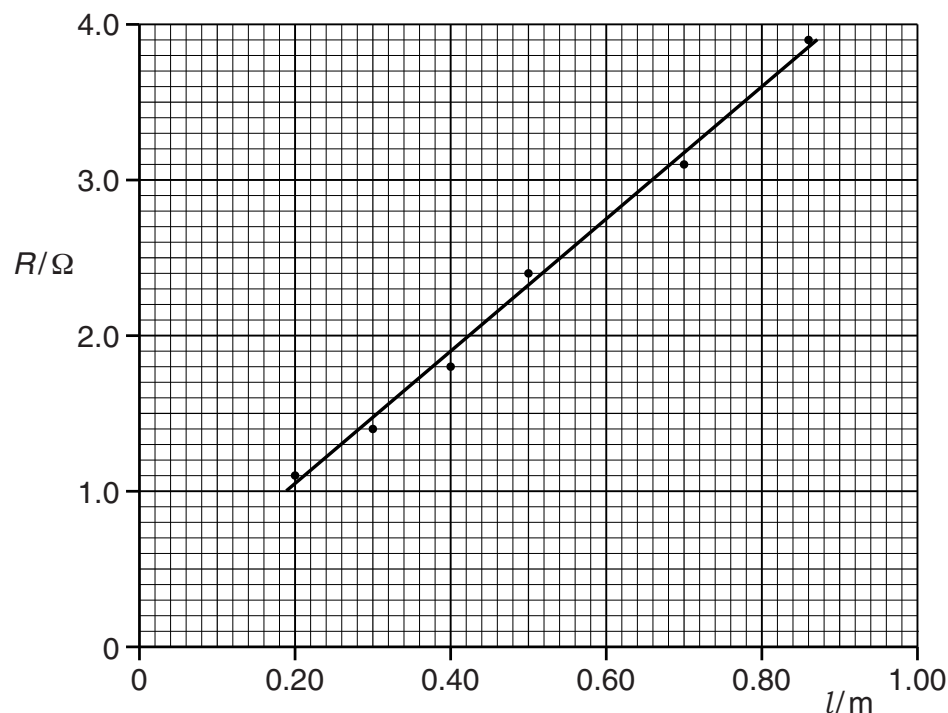


Fig. 3.1

- (a) The points shown in Fig. 3.1 do not lie on the best-fit line. Suggest a reason for this.

.....  
 .....[1]

- (b) Determine the gradient of the line shown in Fig. 3.1.

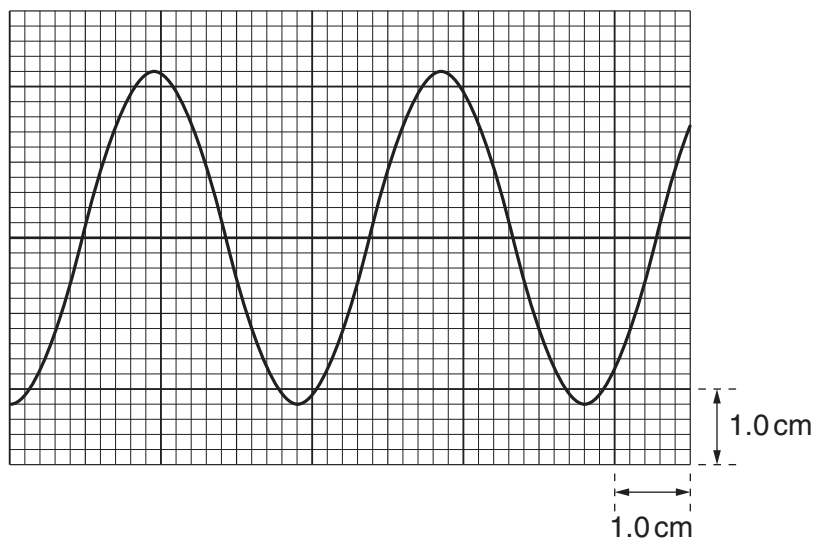
gradient = ..... [2]

- (c) The cross-sectional area of the wire is  $0.12 \text{ mm}^2$ .

Use your answer in (b) to determine the resistivity of the metal of the wire.

resistivity = .....  $\Omega \text{ m}$  [3]

- 40 A microphone detects a musical note of frequency  $f$ . The microphone is connected to a cathode-ray oscilloscope (c.r.o.). The signal from the microphone is observed on the c.r.o. as illustrated in Fig. 2.1.



**Fig. 2.1**

The time-base setting of the c.r.o. is  $0.50 \text{ ms cm}^{-1}$ . The Y-plate setting is  $2.5 \text{ mV cm}^{-1}$ .

- (a) Use Fig. 2.1 to determine

- (i) the amplitude of the signal,

amplitude = ..... mV [2]

- (ii) the frequency  $f$ ,

$f = \dots\dots\dots$  Hz [3]

- (iii) the actual uncertainty in  $f$  caused by reading the scale on the c.r.o.

actual uncertainty = ..... Hz [2]

- (b) State  $f$  with its actual uncertainty.

$f = \dots\dots\dots \pm \dots\dots\dots$  Hz [1]

- 41 Fig. 4.1 shows the values obtained in an experiment to determine the Young modulus  $E$  of a metal in the form of a wire.

| quantity      | value                             | instrument |
|---------------|-----------------------------------|------------|
| diameter $d$  | 0.48 mm                           |            |
| length $l$    | 1.768 m                           |            |
| load $F$      | 5.0 N to 30.0 N<br>in 5.0 N steps |            |
| extension $e$ | 0.25 mm to 1.50 mm                |            |

**Fig. 4.1**

- (a) (i) Complete Fig. 4.1 with the name of an instrument that could be used to measure each of the quantities. [3]

- (ii) Explain why a series of values of  $F$ , each with corresponding extension  $e$ , are measured.

.....  
 .....[1]

- 42 The speed  $v$  of a transverse wave on a uniform string is given by the expression

$$v = \sqrt{\frac{Tl}{m}}$$

where  $T$  is the tension in the string,  $l$  is its length and  $m$  is its mass.

An experiment is performed to determine the speed  $v$  of the wave. The measurements are shown in Fig. 1.1.

| quantity | measurement | uncertainty |
|----------|-------------|-------------|
| $T$      | 1.8N        | $\pm 5\%$   |
| $l$      | 126cm       | $\pm 1\%$   |
| $m$      | 5.1g        | $\pm 2\%$   |

**Fig. 1.1**

- (a) State an appropriate instrument to measure the length  $l$ .

..... [1]

- (b) (i) Use the data in Fig. 1.1 to calculate the speed  $v$ .

$v = \dots\dots\dots \text{ms}^{-1}$  [2]

- (ii) Use your answer in (b)(i) and the data in Fig. 1.1 to determine the value of  $v$ , with its absolute uncertainty, to an appropriate number of significant figures.

$v = \dots\dots\dots \pm \dots\dots\dots \text{ms}^{-1}$  [3]

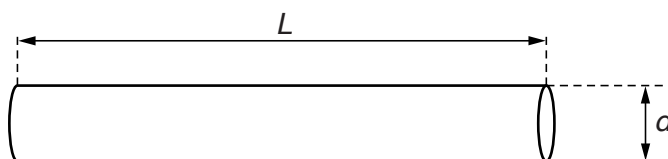
- (i) the mass, in kg, of a wooden metre rule,

mass = ..... kg [1]

- (ii) the volume, in
- $\text{cm}^3$
- , of a cricket ball or a tennis ball.

volume = .....  $\text{cm}^3$  [1]

- (b) A metal wire of length
- $L$
- has a circular cross-section of diameter
- $d$
- , as shown in Fig. 1.1.

**Fig. 1.1**

The volume  $V$  of the wire is given by the expression

$$V = \frac{\pi d^2 L}{4}.$$

The diameter, length and mass  $M$  are measured to determine the density of the metal of the wire. The measured values are:

$$d = 0.38 \pm 0.01 \text{ mm},$$

$$L = 25.0 \pm 0.1 \text{ cm},$$

$$M = 0.225 \pm 0.001 \text{ g}.$$

Calculate the density of the metal, with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

density = .....  $\text{kg m}^{-3}$  [5]

- 44 (a) Describe the effects, one in each case, of systematic errors and random errors when using a micrometer screw gauge to take readings for the diameter of a wire.

systematic errors: .....

.....

random errors: .....

.....[2]

- (b) Distinguish between precision and accuracy when measuring the diameter of a wire.

precision: .....

.....

accuracy: .....

.....[2]

- 45 (a) Define *density*.

.....

.....[1]

- (b) The mass  $m$  of a metal sphere is given by the expression  $m = \frac{\pi d^3 \rho}{6}$

where  $\rho$  is the density of the metal and  $d$  is the diameter of the sphere.

Data for the density and the mass are given in Fig. 1.1.

| quantity | value                    | uncertainty |
|----------|--------------------------|-------------|
| $\rho$   | $8100 \text{ kg m}^{-3}$ | $\pm 5\%$   |
| $m$      | $7.5 \text{ kg}$         | $\pm 4\%$   |

**Fig. 1.1**

- (i) Calculate the diameter  $d$ .

$d = \dots\dots\dots \text{ m [1]}$

- (ii) Use your answer in (i) and the data in Fig. 1.1 to determine the value of  $d$ , with its absolute uncertainty, to an appropriate number of significant figures.

$d = \dots\dots\dots \pm \dots\dots\dots \text{ m [3]}$

- 46 (a) Determine the SI base units of stress.  
Show your working.

9702/21/M/J/17/Q1

base units .....[2]

- (b) A beam PQ is clamped so that the beam is horizontal. A mass  $M$  of 500 g is hung from end Q and the beam bends slightly, as illustrated in Fig. 1.1.

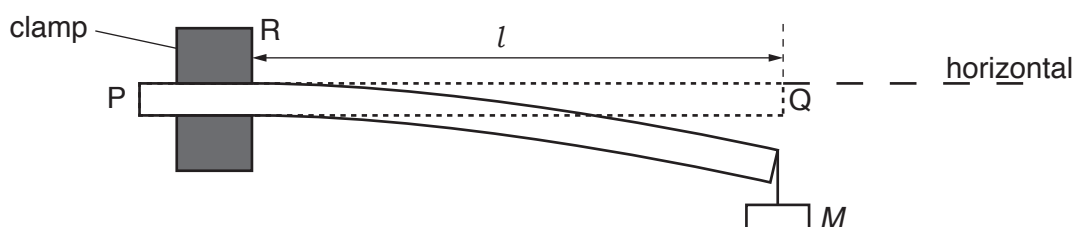


Fig. 1.1

The length  $l$  of the beam from the edge of the clamp R to end Q is 60.0 cm. The width  $b$  of the beam is 30.0 mm and the thickness  $d$  of the beam is 5.00 mm. The material of the beam has Young modulus  $E$ .

The mass  $M$  is made to oscillate vertically. The time period  $T$  of the oscillations is 0.58 s.

The period  $T$  is given by the expression

$$T = 2\pi \sqrt{\frac{4Ml^3}{Ebd^3}}.$$

- (i) Determine  $E$  in GPa.

$E = \dots\dots\dots$  GPa [3]

(ii) The quantities used to determine  $E$  should be measured with accuracy and with precision.

1. Explain the difference between accuracy and precision.

accuracy: .....

.....

precision: .....

.....

[2]

2. In a particular experiment, the quantities  $l$  and  $T$  are measured with the same percentage uncertainty. State and explain which of these two quantities contributes more to the uncertainty in the value of  $E$ .

.....

.....[1]

---

47 1. State what is meant by *precision*.

9702/22/M/J/17/Q1(c)

.....

.....

2. Explain why the precision in the value of the resistivity is improved by using a micrometer screw gauge rather than a metre rule to measure the diameter of the wire.

.....

.....

.....

[2]

---

- 48** One end of a wire is connected to a fixed point. A load is attached to the other end so that the wire hangs vertically.

The diameter  $d$  of the wire and the load  $F$  are measured as  $d = 0.40 \pm 0.02 \text{ mm}$ ,  
 $F = 25.0 \pm 0.5 \text{ N}$ .

- (a)** For the measurement of the diameter of the wire, state

- (i)** the name of a suitable measuring instrument,

.....[1]

- (ii)** how random errors may be reduced when using the instrument in **(i)**.

.....  
 .....[2]

- (b)** The stress  $\sigma$  in the wire is calculated by using the expression  $\sigma = \frac{4F}{\pi d^2}$ .

- (i)** Show that the value of  $\sigma$  is  $1.99 \times 10^8 \text{ N m}^{-2}$ .

[1]

- (ii)** Determine the percentage uncertainty in  $\sigma$ .

percentage uncertainty = .....% [2]

- (iii)** Use the information in **(b)(i)** and your answer in **(b)(ii)** to determine the value of  $\sigma$ , with its absolute uncertainty, to an appropriate number of significant figures.

$\sigma = \dots \pm \dots \text{ N m}^{-2}$  [2]

---

- 49 (a) An analogue voltmeter is used to take measurements of a constant potential difference across a resistor. For these measurements, describe **one** example of

(i) a systematic error, .....  
 .....[1]

(ii) a random error. ....  
 .....[1]

- (b) The potential difference across a resistor is measured as  $5.0\text{ V} \pm 0.1\text{ V}$ . The resistor is labelled as having a resistance of  $125\ \Omega \pm 3\%$ .

- (i) Calculate the power dissipated by the resistor.

power = ..... W [2]

- (ii) Calculate the percentage uncertainty in the calculated power.

percentage uncertainty = ..... % [2]

- (iii) Determine the value of the power, with its absolute uncertainty, to an appropriate number of significant figures.

power = .....  $\pm$  ..... W [2]

---

50 (a) Define *velocity*.

.....  
 .....[1]

(b) The speed  $v$  of a sound wave through a gas of pressure  $P$  and density  $\rho$  is given by the equation

$$v = \sqrt{\frac{kP}{\rho}}$$

where  $k$  is a constant that has no units.

An experiment is performed to determine the value of  $k$ . The data from the experiment are shown in Fig. 1.1.

| quantity | value                              | uncertainty |
|----------|------------------------------------|-------------|
| $v$      | $3.3 \times 10^2 \text{ m s}^{-1}$ | $\pm 3\%$   |
| $P$      | $9.9 \times 10^4 \text{ Pa}$       | $\pm 2\%$   |
| $\rho$   | $1.29 \text{ kg m}^{-3}$           | $\pm 4\%$   |

**Fig. 1.1**

(i) Use data from Fig. 1.1 to calculate  $k$ .

$k = \dots\dots\dots$  [2]

(ii) Use your answer in (b)(i) and data from Fig. 1.1 to determine the value of  $k$ , with its absolute uncertainty, to an appropriate number of significant figures.

$k = \dots\dots\dots \pm \dots\dots\dots$  [3]

- 51 (a) The diameter  $d$  of a cylinder is measured as  $0.0125 \text{ m} \pm 1.6\%$ .

Calculate the absolute uncertainty in this measurement.

absolute uncertainty = ..... m [1]

- (b) The cylinder in (a) stands on a horizontal surface. The pressure  $p$  exerted on the surface by the cylinder is given by

$$p = \frac{4W}{\pi d^2}.$$

The measured weight  $W$  of the cylinder is  $0.38 \text{ N} \pm 2.8\%$ .

- (i) Calculate the pressure  $p$ .

$p = \dots\dots\dots \text{Nm}^{-2}$  [1]

- (ii) Determine the absolute uncertainty in the value of  $p$ .

absolute uncertainty = .....  $\text{Nm}^{-2}$  [2]

---

52 (a) Make estimates of:

(i) the mass, in g, of a new pencil

mass = ..... g [1]

(ii) the wavelength of ultraviolet radiation.

wavelength = ..... m [1]

(b) The period  $T$  of the oscillations of a mass  $m$  suspended from a spring is given by

$$T = 2\pi \sqrt{\frac{m}{k}}$$

where  $k$  is the spring constant of the spring.

The manufacturer of a spring states that it has a spring constant of  $25 \text{ N m}^{-1} \pm 8\%$ . A mass of  $200 \times 10^{-3} \text{ kg} \pm 4 \times 10^{-3} \text{ kg}$  is suspended from the end of the spring and then made to oscillate.

(i) Calculate the period  $T$  of the oscillations.

$T =$  ..... s [1]

(ii) Determine the value of  $T$ , with its absolute uncertainty, to an appropriate number of significant figures.

$T =$  .....  $\pm$  ..... s [3]

---

- 53 (a) Use an expression for work done, in terms of force, to show that the SI base units of energy are  $\text{kg m}^2 \text{s}^{-2}$ .

[2]

- (b) (i) The energy  $E$  stored in an electrical component is given by

$$E = \frac{Q^2}{2C}$$

where  $Q$  is charge and  $C$  is a constant.

Use this equation and the information in (a) to determine the SI base units of  $C$ .

SI base units ..... [2]

- (ii) Measurements of a constant current in a wire are taken using an analogue ammeter.

For these measurements, describe one possible cause of:

1. a random error

.....  
 .....

2. a systematic error.

.....  
 .....

[2]

**54. 9702/23/O/N/20/Q1**

**(a)** An electromagnetic wave has a wavelength of  $85\ \mu\text{m}$ .

**(i)** State the wavelength, in m, of the wave.

wavelength = ..... m [1]

**(ii)** Calculate the frequency, in THz, of the wave.

frequency = ..... THz [2]

**(iii)** State the name of the region of the electromagnetic spectrum that contains this wave.

..... [1]

**(b)** The current  $I$  in a coil of wire produces a magnetic field. The energy  $E$  stored in the magnetic field is given by

$$E = \frac{I^2 L}{2}$$

where  $L$  is a constant.

The manufacturer of the coil states that the value of  $L$ , in SI base units, is  $7.5 \times 10^{-6} \pm 5\%$ .  
The current  $I$  in the coil is measured as  $(0.50 \pm 0.02)\ \text{A}$ .

The values of  $L$  and  $I$  are used to calculate  $E$ .

Determine the percentage uncertainty in the value of  $E$ .

percentage uncertainty = ..... % [2]

[Total: 6]

55. 9702/21/M/J/21/Q1

(a) Define *density*.

.....  
 ..... [1]

(b) Fig. 1.1 shows a solid pyramid with a square base.

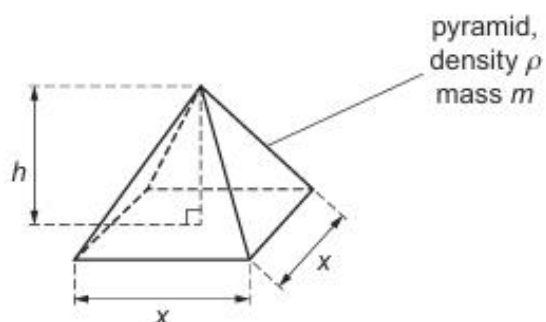


Fig. 1.1

The mass  $m$  of the pyramid is given by

$$m = \frac{1}{3} \rho h x^2$$

where  $\rho$  is the density of the material of the pyramid,  
 $h$  is the height, and  
 $x$  is the length of each side of the base.

Measurements are taken as shown in Table 1.1.

Table 1.1

| quantity | measurement | percentage uncertainty |
|----------|-------------|------------------------|
| $m$      | 19.5 g      | $\pm 2\%$              |
| $x$      | 4.0 cm      | $\pm 5\%$              |
| $h$      | 4.8 cm      | $\pm 4\%$              |

(i) Calculate the absolute uncertainty in length  $x$ .

absolute uncertainty = ..... cm [1]

- (ii) The density  $\rho$  is calculated from the measurements in Table 1.1.

Determine the percentage uncertainty in the calculated value of  $\rho$ .

percentage uncertainty = ..... % [2]

- (c) The square base of the pyramid in (b) rests on the horizontal surface of a bench.

Use data from Table 1.1 to calculate the average pressure of the pyramid on the surface of the bench. The uncertainty in your answer is not required.

pressure = ..... Pa [3]

[Total: 7]

56. 9702/23/M/J/21/Q1

- (a) A property of a vector quantity, that is not a property of a scalar quantity, is direction.  
For example, velocity has direction but speed does not.

- (i) State **two** other scalar quantities and **two** other vector quantities.

scalar quantities: ..... and .....

vector quantities: ..... and .....

[2]

- (ii) State **two** properties that are possessed by both scalar and vector physical quantities.

1. ....

2. ....

[2]

- (b) A ship at sea is travelling with a velocity of  $13 \text{ m s}^{-1}$  in a direction  $35^\circ$  east of north in still water, as shown in Fig. 1.1.

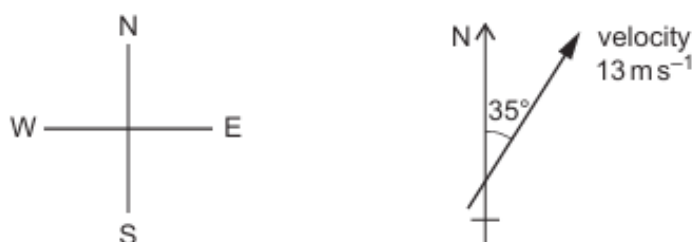


Fig. 1.1

- (i) Determine the magnitudes of the components of the velocity of the ship in the north and the east directions.

north component of velocity = .....  $\text{m s}^{-1}$

east component of velocity = .....  $\text{m s}^{-1}$

[2]

- (ii) The ship now experiences a tidal current. The water in the sea moves with a velocity of  $2.7 \text{ m s}^{-1}$  to the west.

Calculate the resultant velocity component of the ship in the east direction.

resultant east component of velocity = .....  $\text{m s}^{-1}$  [1]

- (iii) Use your answers in (b)(i) and (b)(ii) to determine the magnitude of the resultant velocity of the ship.

magnitude of resultant velocity = .....  $\text{m s}^{-1}$  [2]

- (iv) Use your answers in (b)(i) and (b)(ii) to determine the angle between north and the resultant velocity of the ship.

angle = .....  $^{\circ}$  [2]

[Total: 11]

57. 9702/21/O/N/21/Q1

(a) Define *density*.

.....  
 ..... [1]

(b) A smooth pebble, made from uniform rock, has the shape of an elongated sphere as shown in Fig. 1.1.

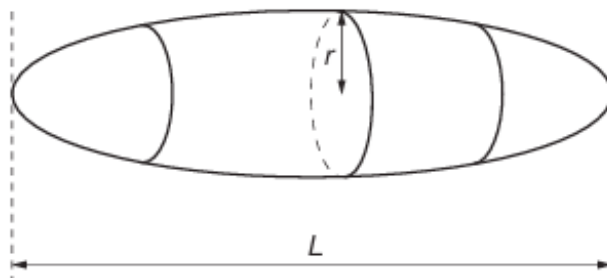


Fig. 1.1

The length of the pebble is  $L$ . The cross-section of the pebble, in the plane perpendicular to  $L$ , is circular with a maximum radius  $r$ .

A student investigating the density of the rock makes measurements to determine the values of  $L$ ,  $r$  and the mass  $M$  of the pebble as follows:

$$\begin{aligned} L &= (0.1242 \pm 0.0001)\text{m} \\ r &= (0.0420 \pm 0.0004)\text{m} \\ M &= (1.072 \pm 0.001)\text{kg}. \end{aligned}$$

(i) State the name of a measuring instrument suitable for making this measurement of  $L$ .

..... [1]

(ii) Determine the percentage uncertainty in the measurement of  $r$ .

percentage uncertainty = ..... % [1]

- (c) The density  $\rho$  of the rock from which the pebble in (b) is composed is given by

$$\rho = \frac{Mr^n}{kL}$$

where  $n$  is an integer and  $k$  is a constant, with no units, that is equal to 2.094.

- (i) Use SI base units to show that  $n$  is equal to  $-2$ .

[2]

- (ii) Calculate the percentage uncertainty in  $\rho$ .

percentage uncertainty = ..... % [3]

- (iii) Determine  $\rho$  with its absolute uncertainty. Give your values to the appropriate number of significant figures.

$\rho = (\text{.....} \pm \text{.....}) \text{ kg m}^{-3}$  [3]

[Total: 11]

**58. 9702/22/O/N/21/Q1**

- (a) A unit may be stated with a prefix that represents a power-of-ten multiple or submultiple.

Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power-of-ten multiple or submultiple.

**Table 1.1**

| prefix   | power-of-ten multiple<br>or submultiple |
|----------|-----------------------------------------|
| kilo (k) | $10^3$                                  |
| tera (T) |                                         |
| ( )      | $10^{-12}$                              |

[2]

- (b) In the following list, underline all the units that are SI base units.

ampere

coulomb

metre

newton

[1]

- (c) The potential difference  $V$  between the two ends of a uniform metal wire is given by

$$V = \frac{4\rho LI}{\pi d^2}$$

where  $d$  is the diameter of the wire,

$I$  is the current in the wire,

$L$  is the length of the wire,

and  $\rho$  is the resistivity of the metal.

For a particular wire, the percentage uncertainties in the values of some of the above quantities are listed in Table 1.2.

**Table 1.2**

| quantity | percentage uncertainty |
|----------|------------------------|
| $d$      | $\pm 3.0\%$            |
| $I$      | $\pm 2.0\%$            |
| $L$      | $\pm 2.5\%$            |
| $V$      | $\pm 3.5\%$            |

The quantities listed in Table 1.2 have values that are used to calculate  $\rho$  as  $4.1 \times 10^{-7} \Omega \text{m}$ .

For this value of  $\rho$ , calculate:

- (i) the percentage uncertainty

percentage uncertainty = .....% [2]

- (ii) the absolute uncertainty.

absolute uncertainty = .....  $\Omega \text{m}$  [1]

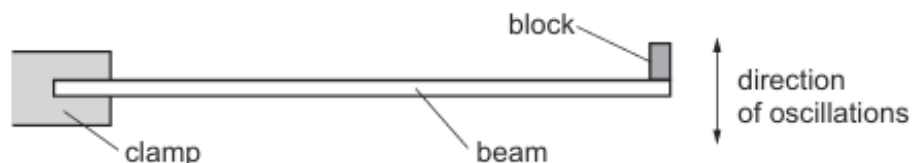
[Total: 6]

**59. 9702/22/M/J/22/Q1**

(a) In the following list, underline all units that are SI base units.

ampere      degree Celsius      kilogram      newton      [1]

(b) Fig. 1.1 shows a horizontal beam clamped at one end with a block attached to the other end.



**Fig. 1.1**

The block is made to oscillate vertically.

The Young modulus  $E$  of the material of the beam is given by

$$E = \frac{kM}{T^2}$$

where  $M$  is the mass of the block,  
 $T$  is the period of the oscillations  
 and  $k$  is a constant.

A student determines the values and percentage uncertainties of  $k$ ,  $M$  and  $T$ .  
 Table 1.1 lists the percentage uncertainties.

**Table 1.1**

| quantity | percentage uncertainty |
|----------|------------------------|
| $k$      | $\pm 2.1\%$            |
| $M$      | $\pm 0.6\%$            |
| $T$      | $\pm 1.5\%$            |

The student uses the values of  $k$ ,  $M$  and  $T$  to calculate the value of  $E$  as  $8.245 \times 10^9 \text{ Pa}$ .

(i) Calculate the percentage uncertainty in the value of  $E$ .

percentage uncertainty = ..... % [2]

- (ii) Use your answer in **(b)(i)** to determine the value of  $E$ , with its absolute uncertainty, to an appropriate number of significant figures.

$$E = (\text{.....} \pm \text{.....}) \times 10^9 \text{ Pa} \quad [2]$$

[Total: 5]

**60. 9702/23/M/J/22/Q1**

A solid metal sphere has a diameter of  $(3.42 \pm 0.02)\text{cm}$  and a mass of  $(67 \pm 2)\text{g}$ .

(a) Calculate the density, in  $\text{g cm}^{-3}$ , of the metal.

density = .....  $\text{g cm}^{-3}$  [3]

(b) Determine the percentage uncertainty in the density.

percentage uncertainty = .....% [2]

[Total: 5]

**61. 9702/21/O/N/22/Q1**

- (a) The boxes in Fig. 1.1 contain terms on the left-hand side and examples of these terms on the right-hand side.

Draw a line between each term on the left and the correct example on the right.



**Fig. 1.1**

[2]

- (b) A set of experimental measurements is described as precise and not accurate.

State what is meant by:

- (i) precise

.....  
..... [1]

- (ii) not accurate.

.....  
..... [1]

- (c) An object of mass  $m$  travels with speed  $v$  in a circle of radius  $r$ . The force  $F$  acting on the object is given by

$$F = \frac{mv^2}{r}.$$

The percentage uncertainties of three of the quantities are given in Table 1.1.

**Table 1.1**

| quantity | percentage uncertainty |
|----------|------------------------|
| $F$      | $\pm 3\%$              |
| $m$      | $\pm 4\%$              |
| $r$      | $\pm 5\%$              |

The value of  $v$  is determined from  $F$ ,  $m$  and  $r$ .

- (i) Calculate the percentage uncertainty in  $v$ .

percentage uncertainty = ..... % [2]

- (ii) The value of  $v$  is  $15.0 \text{ ms}^{-1}$ .

Calculate the absolute uncertainty in  $v$ .

absolute uncertainty = .....  $\text{ms}^{-1}$  [1]

[Total: 7]

**62. 9702/22/O/N/22/Q1**

- (a)** State what is meant by work done.

.....  
..... [1]

- (b)** Use the answer to **(a)** to determine the SI base units of power.

SI base units ..... [2]

- (c)** The maximum useful output power  $P$  of a car travelling on a horizontal road is given by

$$P = v^3 b$$

where  $v$  is the maximum speed of the car and  $b$  is a constant.

For the car,

$$P = 84 \text{ kW} \pm 5\%$$

and  $b = 0.56 \pm 7\%$  in SI units.

- (i)** Calculate the value of  $v$ .

$v = \dots\dots\dots \text{ m s}^{-1}$  [2]

- (ii)** Determine the absolute uncertainty in the value of  $v$ .

absolute uncertainty = .....  $\text{ m s}^{-1}$  [2]

[Total: 7]

**63. 9702/23/O/N/22/Q1**

The rate of flow  $Q$  of a liquid along a narrow pipe of length  $L$  and radius  $r$  is given by

$$Q = \frac{\alpha r^4}{L}$$

where  $\alpha$  is a constant.

An experiment is carried out to determine the value of  $\alpha$ . The data from the experiment are shown in Table 1.1.

**Table 1.1**

| quantity | value                                            | percentage uncertainty |
|----------|--------------------------------------------------|------------------------|
| $Q$      | $2.72 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}$ | $\pm 3\%$              |
| $r$      | $7.1 \times 10^{-5} \text{ m}$                   | $\pm 2\%$              |
| $L$      | $2.5 \times 10^{-2} \text{ m}$                   | $\pm 4\%$              |

- (a) Use information in Table 1.1 to show that the SI base unit of  $\alpha$  is  $\text{s}^{-1}$ .

[1]

- (b) Show that the percentage uncertainty in  $\alpha$  is 15%.

[1]

- (c) Calculate  $\alpha$  with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

$$\alpha = ( \dots \pm \dots ) \times 10^7 \text{ s}^{-1} \quad [3]$$

[Total: 5]

**64. 9702/22/F/M/23/Q1**

- (a) Underline **all** the SI base units in the following list.

ampere                  coulomb                  current                  kelvin                  newton                  [1]

- (b) A toy car moves in a horizontal straight line. The displacement  $s$  of the car is given by the equation

$$s = \frac{v^2}{2a}$$

where  $a$  is the acceleration of the car and  $v$  is its final velocity.

State **two** conditions that apply to the motion of the car in order for the above equation to be valid.

1 .....

2 .....

[2]

- (c) An experiment is performed to determine the acceleration of the car in (b). The following measurements are obtained:

$$s = 3.89 \text{ m} \pm 0.5\%$$

$$v = 2.75 \text{ m s}^{-1} \pm 0.8\%.$$

- (i) Calculate the acceleration  $a$  of the car.

$$a = \dots \text{ m s}^{-2} \quad [1]$$

- (ii) Determine the percentage uncertainty, to two significant figures, in  $a$ .

$$\text{percentage uncertainty} = \dots \% \quad [2]$$

- (iii) Use your answers in (c)(i) and (c)(ii) to determine the absolute uncertainty in the calculated value of  $a$ .

absolute uncertainty = .....  $\text{ms}^{-2}$  [1]

[Total: 7]

**65. 9702/21/M/J/23/Q1**

**(a) (i)** Define power.

.....  
..... [1]

**(ii)** Use the definition of power to show that the SI base units of power are  $\text{kg m}^2 \text{s}^{-3}$ .

[1]

**(b)** The intensity  $I$  of a sound wave moving through a gas is given by

$$I = f^2 A^2 v k$$

where  $f$  is the frequency of the wave,  
 $A$  is the amplitude of the wave,  
 $v$  is the speed of the wave  
and  $k$  is a constant that depends on the gas.

Determine the SI base units of  $k$ .

SI base units ..... [3]

[Total: 5]

**66. 9702/22/M/J/23/Q1**

- (a) (i)** Define pressure.

.....  
 ..... [1]

- (ii)** Use the answer to **(a)(i)** to show that the SI base units of pressure are  $\text{kg m}^{-1} \text{s}^{-2}$ .

[1]

- (b)** A horizontal pipe has length  $L$  and a circular cross-section of radius  $R$ . A liquid of density  $\rho$  flows through the pipe. The mass  $m$  of liquid flowing through the pipe in time  $t$  is given by

$$m = \frac{\pi(p_2 - p_1)R^4 \rho t}{8kL}$$

where  $p_1$  and  $p_2$  are the pressures at the ends of the pipe and  $k$  is a constant.

Determine the SI base units of  $k$ .

SI base units ..... [3]

- (c)** An experiment is performed to determine the value of  $k$  by measuring the values of the other quantities in the equation in **(b)**.

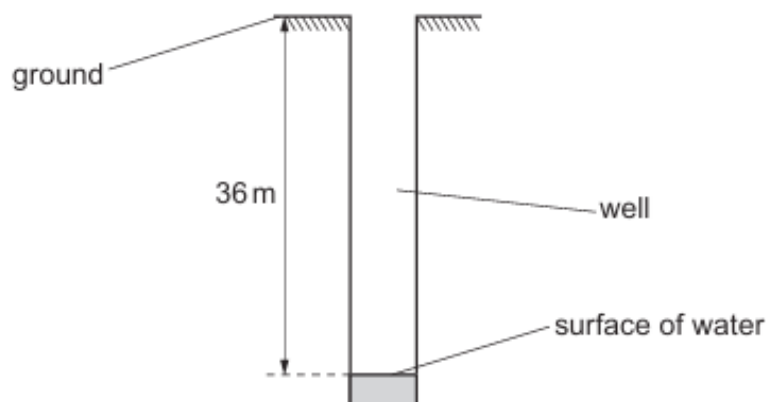
The values of  $L$  and  $R$  each have a percentage uncertainty of 2%.

State and explain, quantitatively, which of these two quantities contributes more to the percentage uncertainty in the calculated value of  $k$ .

.....  
 .....  
 ..... [1]

**67. 9702/23/M/J/23/Q1**

A well has a depth of 36 m from ground level to the surface of the water in the well, as shown in Fig. 1.1.



**Fig. 1.1** (not to scale)

A student wishes to find the depth of the well. The student plans to drop a stone down the well and record the time taken from releasing the stone to hearing the splash made by the stone as it enters the water.

- (a) Assume that air resistance is negligible and that the stone is released from rest.

Calculate the time taken for the stone to fall from ground level to the surface of the water.

time = ..... s [2]

- (b) The time recorded by the student using a stop-watch is not equal to the time in (a).

Suggest **three** possible reasons, other than the effect of air resistance, for this difference.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

- (c) The student repeats the experiment three times and uses the results to calculate the depth of the well. The values are shown in Table 1.1.

**Table 1.1**

|         | 1st experiment | 2nd experiment | 3rd experiment |
|---------|----------------|----------------|----------------|
| depth/m | 54.4           | 53.9           | 54.1           |

The true depth of the well is 36.0 m. Explain why these results may be described as precise but not accurate.

.....

.....

.....

..... [2]

[Total: 7]

68. 9702/21/O/N/23/Q1

- (a) Compare scalar and vector quantities.

.....

.....

..... [2]

- (b) The radius of a small sphere is determined from a measurement of the volume of the sphere. The sphere is submerged in water, displacing some of the water into a measuring cylinder as shown in Fig. 1.1.

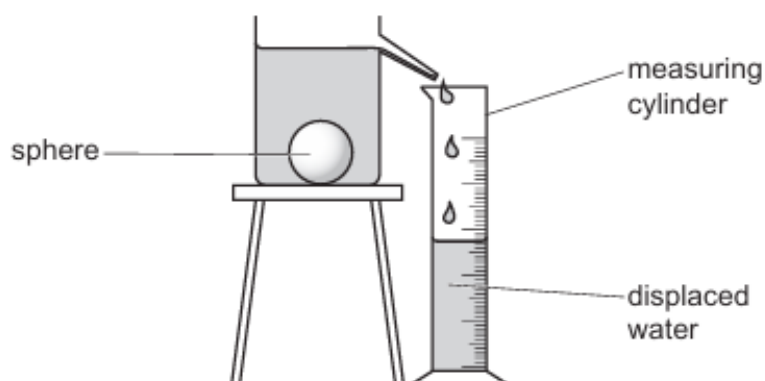


Fig. 1.1 (not to scale)

The measured volume of displaced water is  $(28.0 \pm 0.5) \text{ cm}^3$ .

Calculate:

- (i) the radius, in cm, of the sphere

radius = ..... cm [1]

- (ii) the percentage uncertainty in the radius of the sphere.

percentage uncertainty = ..... % [2]

[Total: 5]

**69. 9702/23/O/N/23/Q1**

- (a) Table 1.1 lists some physical quantities. Complete the table by placing a tick (✓) next to the scalar quantities.

**Table 1.1**

|              |  |
|--------------|--|
| acceleration |  |
| charge       |  |
| momentum     |  |
| power        |  |
| upthrust     |  |

[1]

- (b) A uniform cylinder has diameter  $D$ , length  $L$  and mass  $M$ .  
The density  $\rho$  of the cylinder is given by

$$\rho = \frac{4M}{\pi D^2 L}.$$

Table 1.2 shows the data obtained from an experiment to determine the density of the cylinder.

**Table 1.2**

| quantity | measurement               | percentage uncertainty |
|----------|---------------------------|------------------------|
| $D$      | $(26.2 \pm 0.1)\text{mm}$ | .....%                 |
| $L$      | $(162 \pm 1)\text{mm}$    | .....%                 |
| $M$      | $(247 \pm 1)\text{g}$     | 0.4%                   |

- (i) Calculate the percentage uncertainties in  $D$  and  $L$ . Write your answers in Table 1.2.

[1]

(ii) Calculate the density of the cylinder. Give your answer to three significant figures.

density = .....  $\text{kg m}^{-3}$  [2]

(iii) Calculate the percentage uncertainty in the density.

percentage uncertainty = .....% [2]

[Total: 6]